

19 Capacitance

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Capacitance $C = Q/V$ measures charge stored per potential difference. A capacitor stores energy $E = \frac{1}{2}QV = \frac{1}{2}CV^2$. During resistor-capacitor discharge, $V = V_0e^{(-t/RC)}$; the time constant RC is the time for voltage to fall to $1/e$ of its initial value.

Worked example: follow the reasoning

For $C = 100 \mu\text{F}$ at 12 V , $Q = 1.2 \times 10^{-3} \text{ C}$ and stored energy $= \frac{1}{2} \times 100 \times 10^{-6} \times 12^2 = 7.2 \text{ mJ}$.

Quick check · multiple choice

A For capacitance C at p.d. V , energy stored is what?

[1]

A CV^2

B $2CV^2$

C $\frac{1}{2}CV^2$

D $\frac{1}{2}C/V^2$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A $220 \mu\text{F}$ capacitor discharges through $10 \text{ k}\Omega$. Find its time constant and fraction of voltage remaining after one time constant.

[3]

Final answer: _____

Answers and explanation

Quick check: C. $\frac{1}{2}CV^2$

Question 1 [3 marks]

$\tau = RC = 10\,000 \times 220 \times 10^{-6} = 2.2 \text{ s}$ [2]; fraction $e^{-1} \approx 0.368$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.